US012404154B2

(12) **United States Patent**
Honjo et al.

(10) **Patent No.: US 12,404,154 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **WORK MACHINE AND MOBILE CRANE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 372 days.

(21) Appl. No.: **17/951,890**

(22) Filed: **Sep. 23, 2022**

(65) **Prior Publication Data**

US 2023/0018143 A1 Jan. 19, 2023

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2021/012987, filed on Mar. 26, 2021.

(30) **Foreign Application Priority Data**

Mar. 27, 2020 (JP) 2020-057213

(51) **Int. Cl.**
B66C 23/74 (2006.01)
B66C 13/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B66C 23/74*** (2013.01); ***B66C 13/00*** (2013.01); ***B66C 23/88*** (2013.01); *B66C 23/36* (2013.01); *B66C 2700/0371* (2013.01)

(58) **Field of Classification Search**
CPC B66C 23/72; B66C 23/74; B66C 23/76; E02F 9/18; B62D 49/085
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,524,777	A *	6/1996	Weber	B66C 23/74 212/178
10,717,631	B2	7/2020	Rudy et al.	

(Continued)

FOREIGN PATENT DOCUMENTS

CN	106144924	A	11/2016
JP	H07-117989	A	5/1995

(Continued)

OTHER PUBLICATIONS

International Search Report issued in Application No. PCT/JP2021/012987, mailed Jun. 22, 2021.

(Continued)

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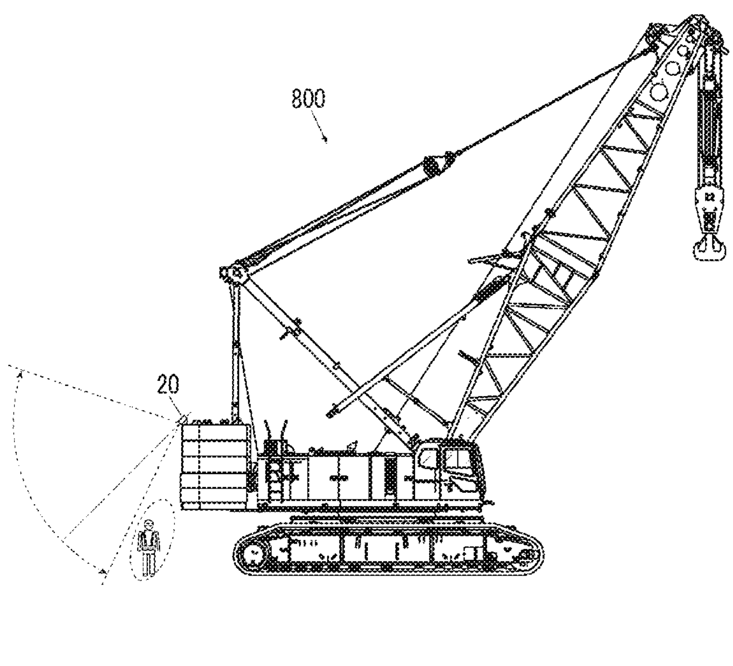
(57) **ABSTRACT**

The work machine includes a work machine body, and an assembly component that is attached to the work machine body. The work machine further includes a detection unit that monitors surroundings. The work machine body includes a lower traveling body and a rotating platform that rotates relative to the lower traveling body. The detection unit is attached to the rotating platform without interposing the assembly component.

10 Claims, 7 Drawing Sheets



(51) **Int. Cl.**
B66C 23/88 (2006.01)
B66C 23/36 (2006.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,829,347 B2	11/2020	Rudy et al.	
10,919,584 B2	2/2021	Takahashi et al.	
11,124,392 B2	9/2021	Rudy et al.	
11,130,658 B2	9/2021	Rudy et al.	
2011/0031202 A1	2/2011	Pech et al.	
2013/0098860 A1	4/2013	Pech et al.	
2013/0112642 A1	5/2013	Pech et al.	
2018/0141787 A1*	5/2018	Rudy	B66C 15/06
2018/0143011 A1*	5/2018	Rudy	G06T 7/70
2018/0338087 A1*	11/2018	Oblak	G06F 3/04883
2020/0062558 A1	2/2020	Pech et al.	
2022/0119232 A1	4/2022	Pech et al.	

FOREIGN PATENT DOCUMENTS

JP	H9-158258 A	6/1997
JP	2000-191273 A	7/2000
JP	2002-128463 A	5/2002
JP	2002-371595 A	12/2002
JP	2006-037606 A	2/2006
JP	2010-189869 A	9/2010
JP	2013-204411 A	10/2013
JP	2018-093501 A	6/2018
JP	2018-127357 A	8/2018
JP	2019-014585 A	1/2019
JP	2019-075712 A	5/2019

OTHER PUBLICATIONS

Chinese Office Action dated Oct. 18, 2024 for the corresponding Chinese Application No. 202180023889.3.

* cited by examiner

FIG. 1

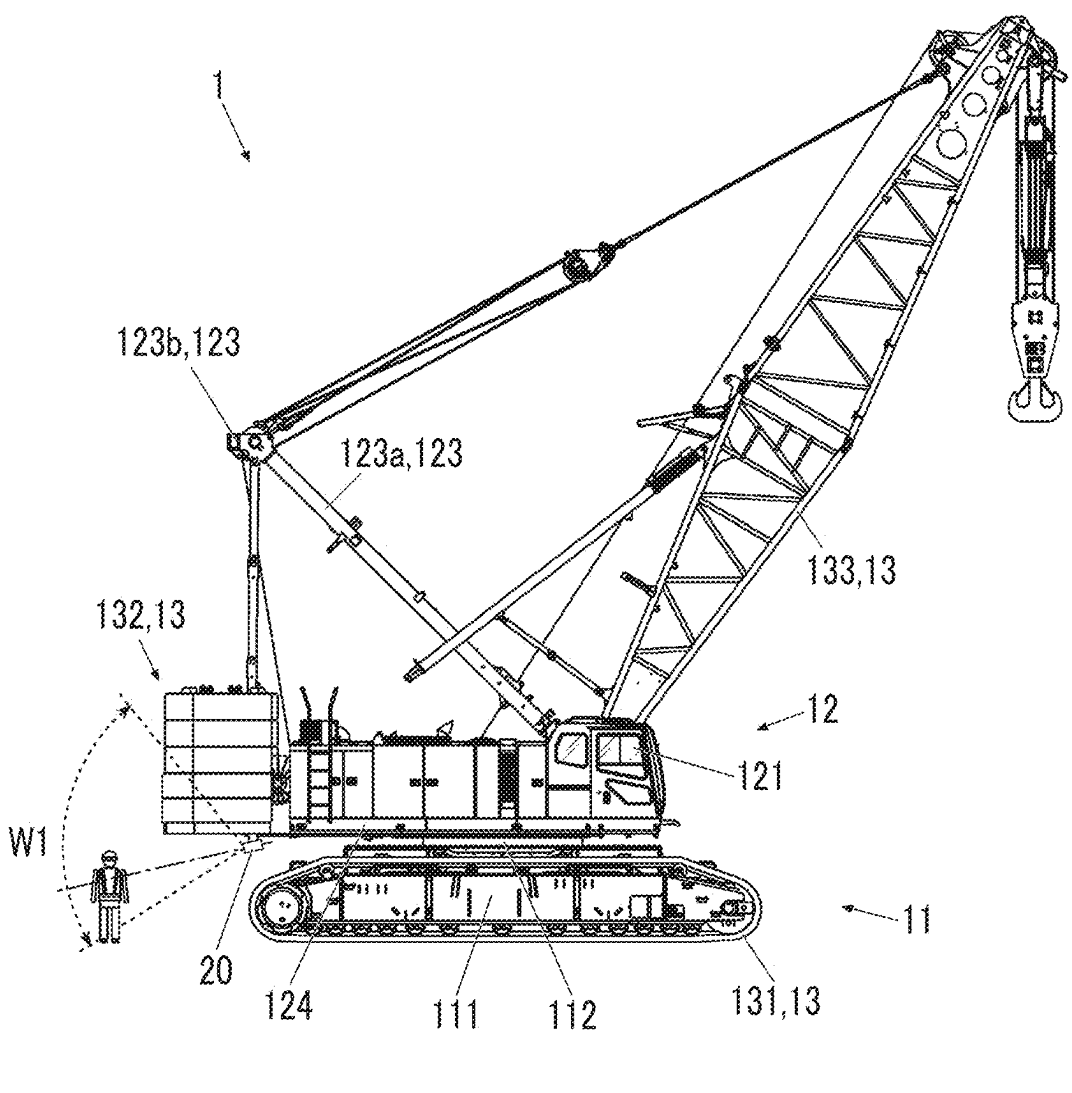
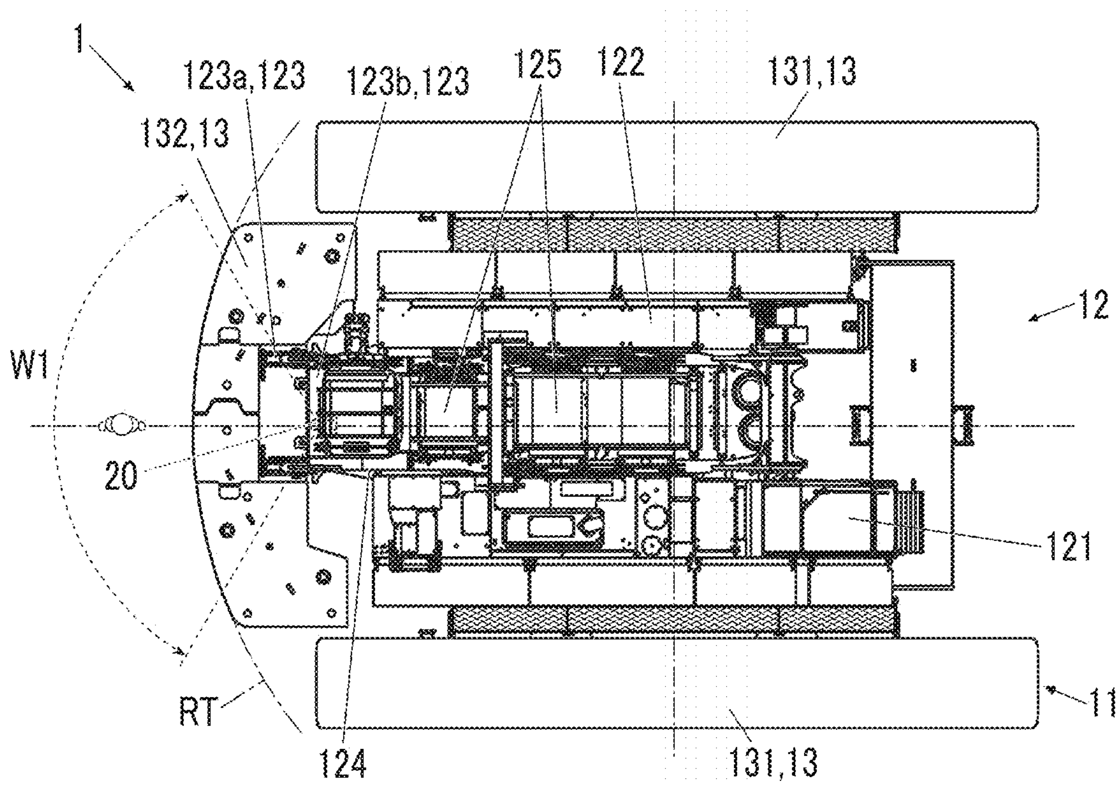

1
123b,123
123a,123
133,13
132,13
12
121
W1
11
20
124
111
112
131,13

FIG. 2



1
123a,123
123b,123
125
122
131,13
132,13
12
W1
20
121
RT
124
131,13
11

FIG. 3

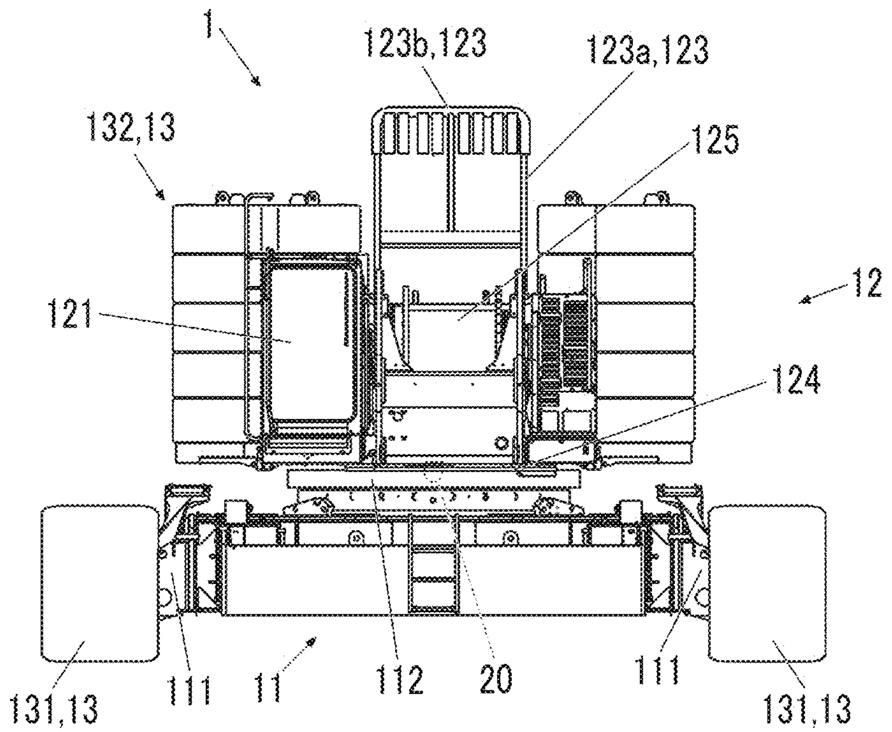

1
123b,123
123a,123
132,13
125
12
121
124
131,13
111
11
112
20
111
131,13

FIG. 4

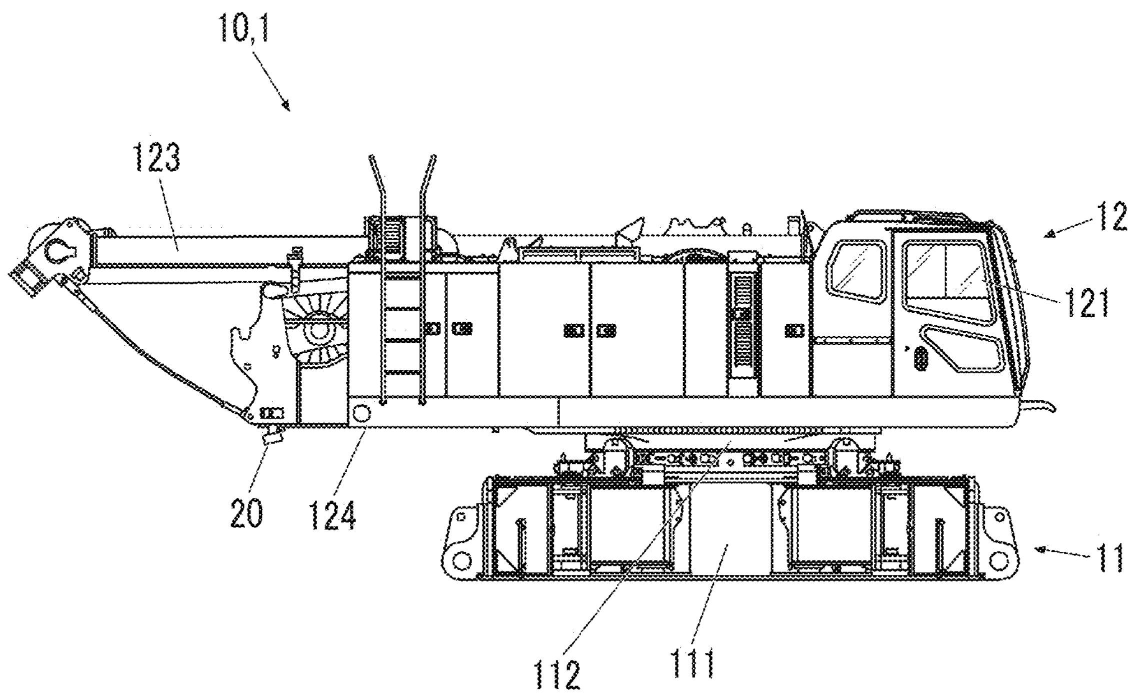

10,1
123
12
121
20
124
11
112
111

FIG. 5A

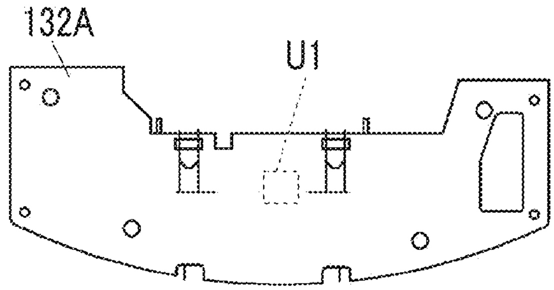


FIG. 5B

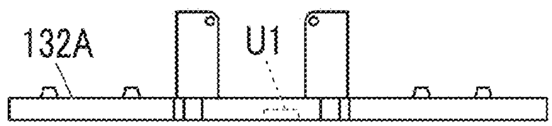


FIG. 5C

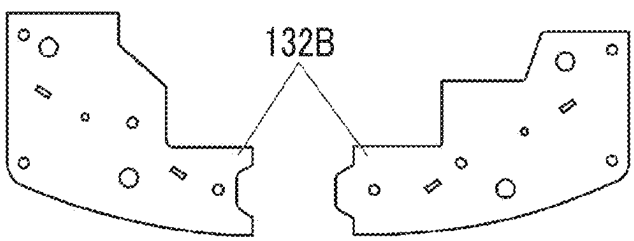


FIG. 5D

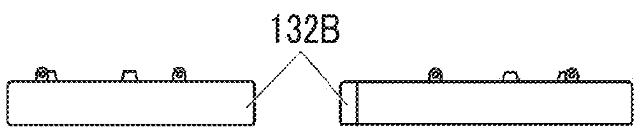


FIG. 5E

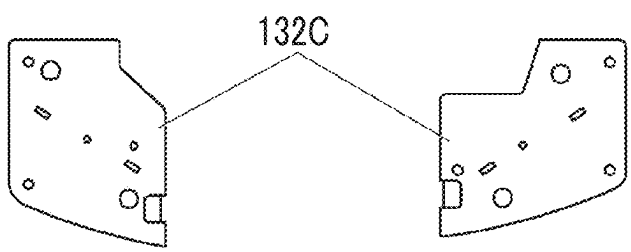


FIG. 5F

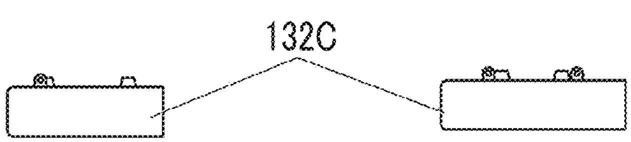


FIG. 6A

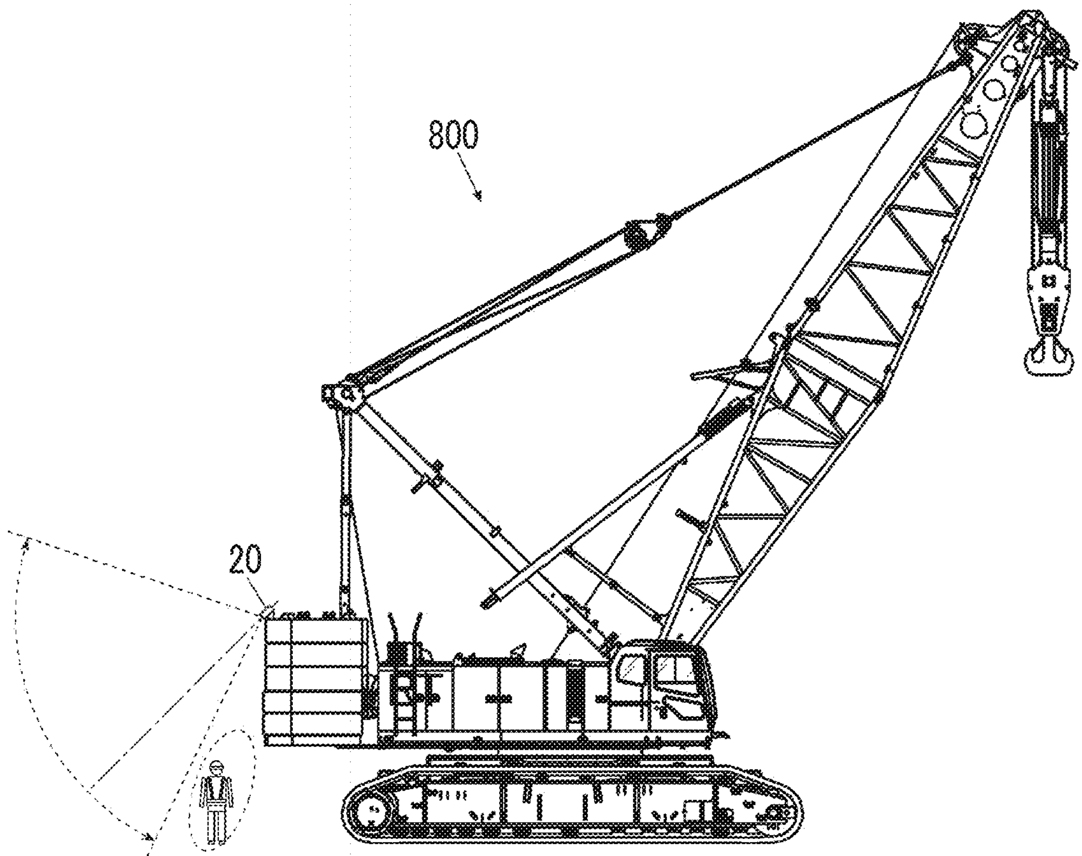


FIG. 6B

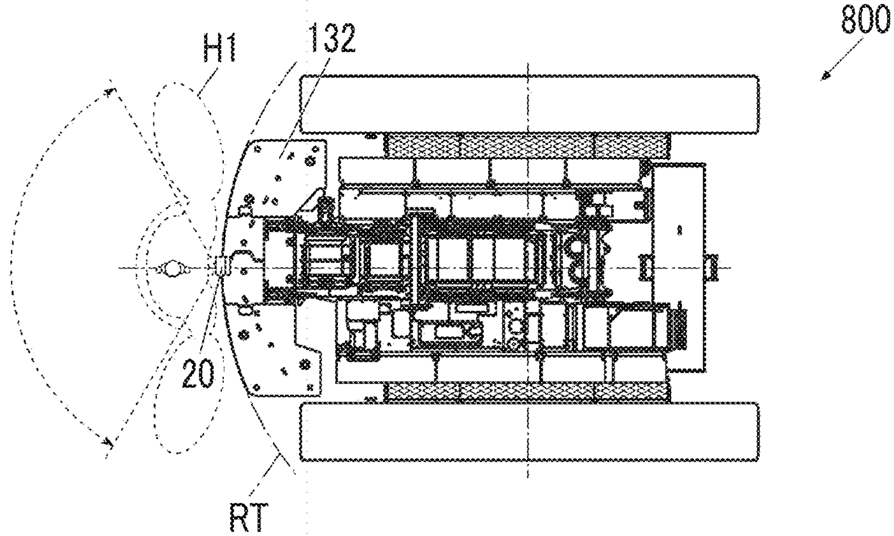


FIG. 7

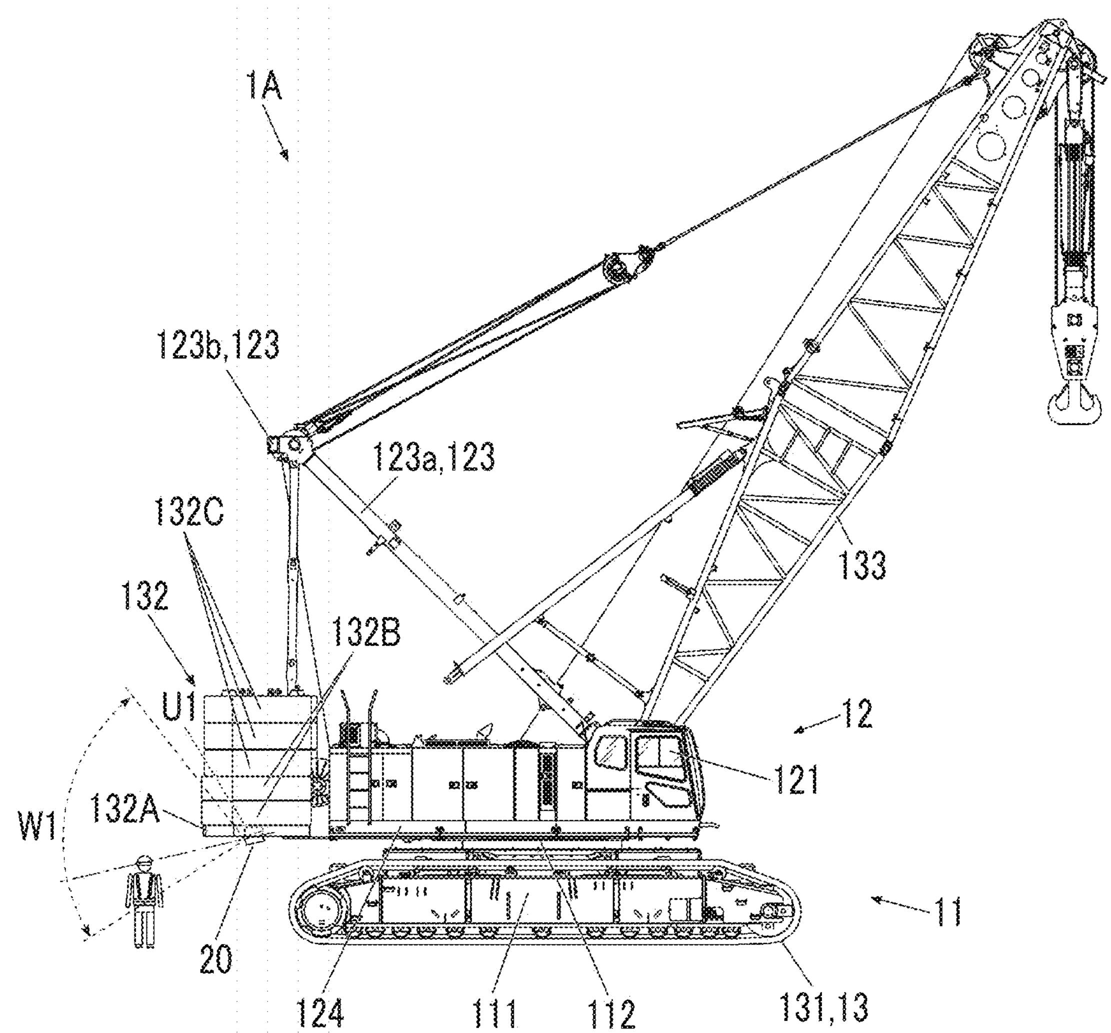
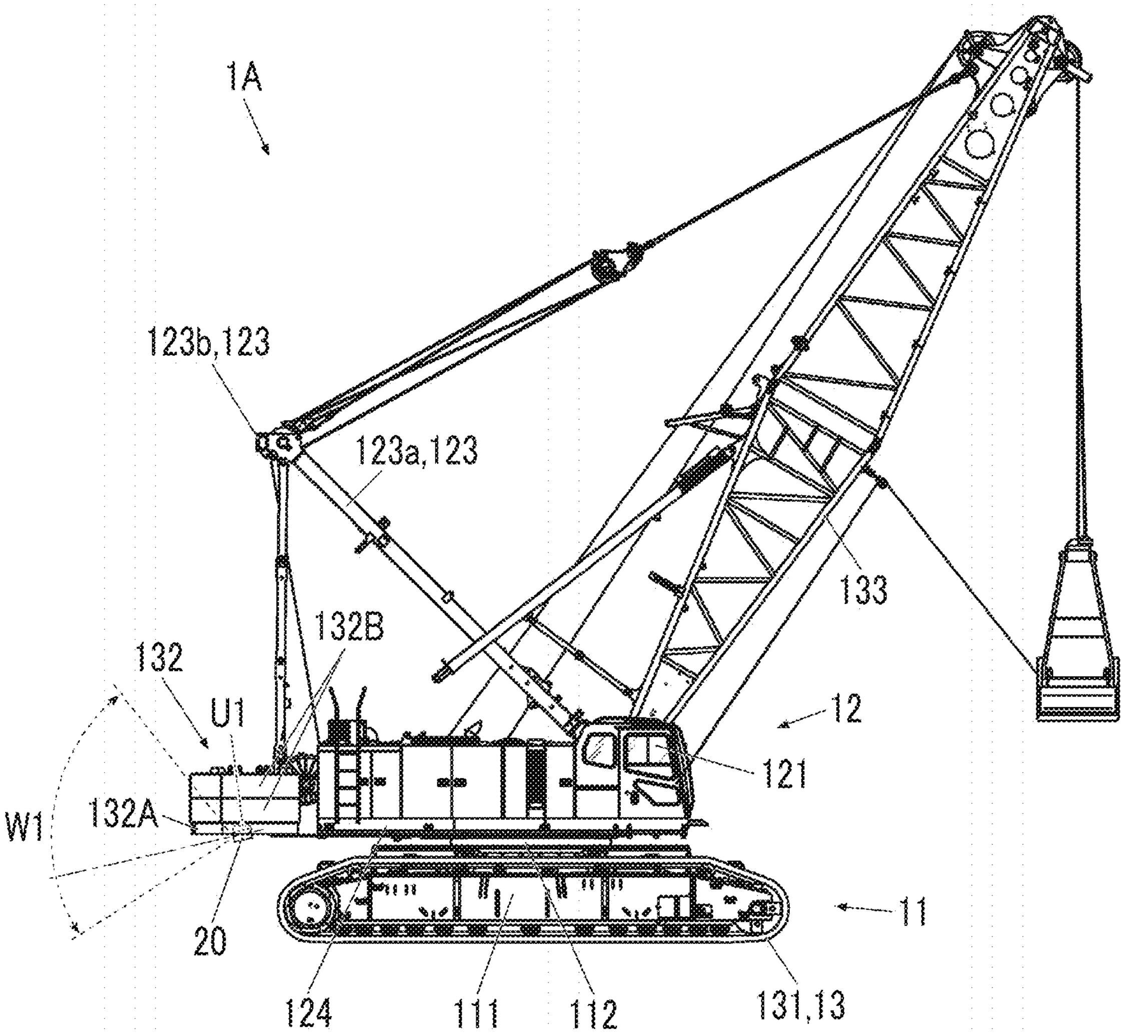

1A
123b,123
123a,123
132C
132
132B
U1
132A
W1
20
124
111
112
131,13
133
12
121
11

FIG. 8


1A
123b,123
123a,123
133
132B
132
U1
12
121
W1
132A
20
11
124
111
112
131,13

WORK MACHINE AND MOBILE CRANE

RELATED APPLICATIONS

The contents of Japanese Patent Application No. 2020-057213, and of International Patent Application No. PCT/JP2021/012987, on the basis of each of which priority benefits are claimed in an accompanying application data sheet, are in their entirety incorporated herein by reference.

BACKGROUND

Technical Field

Certain embodiments of the present invention relate to a work machine and a mobile crane.

Description of Related Art

The related art discloses a surrounding monitoring device for an excavator, which is a work machine. The surrounding monitoring device includes a camera and a distance image sensor attached at an upper rear end of a counterweight of the excavator.

SUMMARY

According to an embodiment of the present invention, there is provided a work machine that is assembled or disassembled, the work machine including:

a work machine body; and

an assembly component that is attached to the work machine body, the work machine further including:

a detection unit that monitors surroundings,

in which the work machine body includes a lower traveling body and a rotating platform that rotates relative to the lower traveling body, and

the detection unit is attached to the rotating platform without interposing the assembly component.

According to another embodiment of the present invention, there is provided a mobile crane that is assembled or disassembled at a work site, the mobile crane including:

a crane body; and

a counterweight that includes weights in a plurality of stages and that is attached to the crane body,

in which a setting is capable of being changed to any one of a plurality of lifting capacities by changing the number of stages of the weights attached to the crane body, and

the plurality of lifting capacities includes a first lifting capacity and a second lifting capacity that is a lower lifting capacity than the first lifting capacity,

the mobile crane further including:

a detection unit that monitors surroundings attached to the counterweight,

in which the detection unit is attached below an uppermost stage of the weight attached when the second lifting capacity is set.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a side view showing a work machine according to Embodiment 1 of the present invention.

FIG. **2** is a plan view of the work machine illustrating the disposition of a detection unit.

FIG. **3** is a front view of the work machine illustrating the disposition of the detection unit.

FIG. **4** is a view showing a configuration of the work machine of FIG. **1** with an assembly component removed.

FIG. **5**A is a view showing a counterweight and a plan view of a base weight.

FIG. **5**B is a view showing the counterweight and a rear view of the base weight.

FIG. **5**C is a view showing the counterweight and a plan view of a first weight.

FIG. **5**D is a view showing the counterweight and a rear view of the first weight.

FIG. **5**E is a view showing the counterweight and a plan view of a second weight.

FIG. **5**F is a view showing the counterweight and a rear view of the second weight.

FIG. **6**A is a side view showing a work machine of a comparative example.

FIG. **6**B is a plan view showing the work machine of the comparative example.

FIG. **7** is a side view showing a state in which a work machine according to Embodiment 2 of the present invention is set to have a first lifting capacity.

FIG. **8** is a side view showing a state in which the work machine according to Embodiment 2 is set to have a second lifting capacity lower than the first lifting capacity.

DETAILED DESCRIPTION

A work machine such as a crawler crane is assembled at a work site, and when the work is completed, the work machine is disassembled at the work site and carried out. Moreover, there is an assembly component that may or may not be attached at each site. In the case of the work machine that is assembled or disassembled in this way, it is necessary to change the position of a surrounding monitoring device depending on the mounting situation of the assembly component, and there is a need in that it is difficult to realize uniform monitoring processing is difficult.

It is desirable to provide a work machine and a mobile crane that are assembled or disassembled and that can realize monitoring processing that is less dependent on the mounting situation of an assembly component.

According to the present invention, in the work machine and the mobile crane that are assembled or disassembled at a work site, surrounding monitoring can be performed even during assembly or disassembly without the need to move a surrounding monitoring device.

Hereinafter, respective embodiments of the present invention will be described in detail with reference to the drawings.

Embodiment 1

FIG. **1** is a side view showing a work machine according to Embodiment 1 of the present invention. FIG. **2** is a plan view of the work machine illustrating the disposition of a detection unit. FIG. **3** is a front view of the work machine illustrating the disposition of the detection unit. FIGS. **2** and **3** show a state in which a boom **133** is removed. FIG. **4** is a view showing a configuration of the work machine of FIG. **1** with an assembly component removed. Hereinafter, a description will be made with a direction along a pivot axis of the boom **133** as a left-right direction, and out of directions perpendicular and horizontal to the pivot axis of the boom **133**, a cab **121** side from a turning center of the rotating platform **12** is described as a front side and a counterweight **132** side is described as a rear side.

A work machine **1** according to Embodiment 1 is a machine that is assembled and used at a work site, or disassembled and transported from the work site. The work machine **1** is a mobile crane, specifically a crawler crane. In addition, the assembly or disassembly may be performed at a place for assembly around a place other than where lifting work or the like is actually performed. For example, the work machine **1** may be assembled halfway up a mountain, and the assembled work machine **1** may travel and move to a work place. In this case, the assembly place is halfway up the mountain.

The work machine **1** includes a work machine body (crane body) **10** (see FIG. **4**), an assembly component **13** that is lifted by a heavy machine and that is attached to the work machine body **10**, and a detection unit **20** that monitors the surroundings. The work machine body **10** in FIG. **4** shows a state in which the assembly component **13** is not attached. The work machine **1** may further include a monitoring device that receives a detection signal from the detection unit **20** and that monitors the surroundings. The monitoring device may include a display unit that outputs a monitoring image including surrounding workers, pedestrians, or obstacles detected by the detection unit **20**. The display unit may be disposed in a cab of the work machine **1**, a management terminal at the work site, a management center away from the site, or the like.

The work machine body **10** includes a lower traveling body **11** to which a crawler component **131** is assembled to constitute a crawler, and a rotating platform **12** that rotates relative to the lower traveling body **11**. The rotating platform **12** includes a cab **121**, a winch **125**, a hydraulic device **122**, a mast **123**, a main frame **124**, a lower frame **111**, which is a support base for the lower traveling body **11**, and the like. The main frame **124** is a member that supports any one of the boom, the hydraulic device, the winch, or the mast from below. Specifically, the main frame **124** is a frame that supports each component of the rotating platform **12** and to which the boom is pivotably connected, is connected to the lower traveling body **11** via a turning bearing, and extends from the cab **121** to a halfway spot of the counterweight **132** in a front-rear direction. The main frame **124** may be configured so that the main frame **124** can be divided and assembled into a plurality of frames such as a front frame and a back frame. Even in this case, the front frame and the back frame are included in the work machine body **10** and are not included in the assembly component **13**. The work machine body **10** has a jack (not shown) for raising itself. The mast **123** is configured to apply a rearward load to the boom **133**. When the work machine **1** is assembled or disassembled, a hook is hung from a tip of the mast **123**, the assembly component **13** is lifted and lowered by the hook and the rotating platform **12** is turned, so that the assembly component **13** lifted by the mast **123** can be moved. The mast **123** is a concept that includes an A-frame that does not oscillate during crane work.

The assembly component **13** is a part that is mainly assembled at a work site, and means a heavy part that is attachable to the work machine body **10** by being lifted by the heavy machine or the like. Therefore, parts such as a lightweight ladder that can be raised and attached only by human power, parts such as a lightweight stay, and a bracket for attaching a camera are excluded from the assembly component **13**. That is, in the present specification, the "assembly component" is defined as a part that cannot be assembled by human power alone. A part that cannot be assembled by human power alone is a part that needs to be lifted by the heavy machine or the like defined in each site or assembly manual, and is, for example, a part larger than 20 kg.

The assembly component **13** includes the crawler component **131**, the counterweight **132**, the boom **133**, a lower weight (not shown), and the like. The crawler component **131** may include a part of a crawler drive mechanism in addition to crawler shoes. The assembly component **13** is connected to the work machine body **10** by pin joints, cylinder fastening, or bolt fastening.

The assembly component **13** is attached at the work site as follows. For example, the crawler component **131** is lifted using another crane or the mast **123** and is moved to the lower frame **111** of the lower traveling body **11** in a state in which the work machine body **10** is raised by the jack. In this state, the worker connects the crawler component **131** to the lower frame of the lower traveling body **11**. The counterweight **132** is lifted using another crane or the mast **123**, moved to a rear portion of the main frame **124**, and attached to a relevant spot. The boom **133** and the lower weights **134** are also lifted using another crane or the mast **123**, moved to an attachment spot of the main frame **124**, and attached to the relevant spot.

FIGS. **5**A to **5**F are views showing the counterweight, FIGS. **5**A, **5**C, and **5**E show plan views of a base weight, a first weight, and a second weight, respectively, and FIGS. **5**B, **5**D, and **5**F show rear views of the base weight, the first weight, and the second weight, respectively. The counterweight **132** includes a base weight **132**A attached to the main frame of the rotating platform **12**, and a first weight **132**B and a second weight **132**C attached to the rotating platform **12** via the base weight **132**A. The base weight **132**A is configured to be attachable only in one stage, and the first weight **132**B and the second weight **132**C of the same form are configured to be stacked in a plurality of stages and be attachable. The counterweight **132** may be configured such that other weights such as the first weight **132**B and the second weight **132**C are stacked above the base weight **132**A or other weights are suspended below the base weight **132**A.

The detection unit **20** is a device that acquires data for detecting objects present in the surroundings, such as a range finder such as a scanning type light detection and ranging (LiDAR) device or an imaging camera. The detection unit **20** has a detection range W**1** shown in FIGS. **1** and **3** and detects an object present within the detection range W**1**. The detection range W**1** extends in an angle range centered on a vertical axis and an angle range centered on an axis extending to the left and right. In a case where the detection unit **20** is a LiDAR device, the detection unit **20** can acquire data indicating the direction and distance of an object located in the detection range and detect where the object is present in the surroundings from the data. In a case where the detection unit **20** is an imaging camera, the detection unit **20** can detect objects such as workers, pedestrians, and obstacles within the detection range W**1** by performing image recognition from the captured image in the detection range W**1**.

As shown in FIG. **1**, the detection unit **20** of the monitoring device is attached to the rotating platform **12** without interposing the assembly component **13**. More specifically, the detection unit **20** is attached to a lower surface of the main frame **124** via a stay or the like and is disposed at a lower portion of the main frame **124** or below the main frame **124**. In addition, the main frame **124** may have a recessed portion at a lower portion thereof, and the detection unit **20** may be disposed in the recessed portion and be positioned above a bottom end of the main frame **124**.

Moreover, the detection unit **20** is attached to a main body of the rotating platform **12** that is first attached to the lower traveling body **11**. However, the attachment of the main body of the rotating platform **12** to the lower traveling body **11** is not performed on site or in its vicinity. The above main body that is first attached to the lower traveling body **11** is, for example, the main frame **124**. Moreover, the detection unit **20** is disposed behind a turning bearing **112** of the rotating platform **12**. In addition, the detection unit **20** is disposed at a position overlapping the mast **123** when viewed vertically from above. The mast **123** has two struts **123***a*, and a horizontal member **123***b* that connects the two struts to each other. The position overlapping the mast **123** when viewed from above may be a position overlapping the strut **123***a* or the horizontal member **123***b* when the mast **123** is at any angle from a horizontal angle to a vertical angle. The position where the mast **123** overlaps corresponds approximately to a center range in the left-right direction behind the center of the rotating platform **12**.

Moreover, the detection unit **20** is disposed inside a turning radius RT (see FIG. **2**) of the rotating platform **12**. The turning radius RT means the trajectory of an end portion of a component that moves on the outermost side when the rotating platform **12** has turned, except for displaceable configurations such as the boom **133** and the mast **123**. In a general crane, the turning radius corresponds to a rear end radius, and corresponds to the trajectory of a rear end portion of the counterweight **132** in a case where the counterweight **132** is attached.

As shown in FIGS. **1** and **3**, the detection range W**1** of the detection unit **20** extends rearward, leftward, rightward, upward, and downward. The detection range W**1** overlaps a part of the counterweight **132**, more specifically, overlaps a lower rear end portion of the counterweight **132**. Therefore, the detection unit **20** can directly detect a portion that forms a turning radius (rear end radius) R**1** of the rotating platform **12**.

FIG. **6**A is a side view showing a work machine of a comparative example. FIG. **6**B is a plan view showing the work machine of the comparative example. In a case where the detection unit **20** is disposed above an upper rear end portion of the counterweight **132** as in a work machine **800** of the comparative example, unless the detection unit **20** is moved to another place before the counterweight **132** is attached, that is, when the work machine **800** is assembled or disassembled, there is a problem that surrounding monitoring cannot be performed. Moreover, since simply moving the detection unit **20** to another position lowers the accuracy of the position and angle of the detection unit **20**, a problem occurs in that favorable surrounding monitoring is difficult.

Additionally, general mobile cranes are configured such that a desired lifting capacity can be selected from a plurality of levels of lifting capacity. The lifting capacity is switched by the number of attachment stages of the counterweights **132**. The number of attachment stages of the counterweights **132** is detected by a weight detection device and notification is sent to the control unit of the work machine, and on the basis of this notification, the control unit sets the lifting capacity and determines various control parameters. When the detection unit **20** is disposed above the upper rear end portion of the counterweight **132**, and when the lifting capacity has been changed, the number of stages of the counterweights **132** changes, and the height of the detection unit **20** changes. When the position of the detection unit **20** is constant, uniform monitoring processing is possible on the basis of the output of the detection unit **20**. However, when the position of the detection unit **20** is not constant, it is necessary to prepare a plurality of kinds of setting data or programs for monitoring processing respectively adapted to a plurality of positions where the detection unit **20** is disposed. When the same monitoring processing is performed using one set data or program even when the position of the detection unit **20** changes, it is difficult to perform the monitoring processing that matches a difference in height.

Additionally, in recent mobile cranes, in order to reduce the size of the crane body portion and to improve the lifting capacity, the weight of the counterweight **132** is increased, and to that effect, the height of the counterweight **132** is often increased. When the detection unit **20** is disposed above the upper rear end portion of the counterweight **132** having an increased height, a blind spot H**1** occurs near the turning radius RT as shown in FIGS. **6**A and **6**B.

Moreover, a fallen mast **123** may approach an upper portion of the counterweight **132**. When the detection unit **20** is disposed in the center such that a single detection unit **20** can monitor a region that extends to the left and right, a risk occurs that the detection unit **20** and the mast **123** interfere with each other.

However, according to the work machine **1** of Embodiment 1, the detection unit **20** for the surrounding monitoring is attached to the rotating platform **12** without interposing the assembly component **13** that is lifted and attached by the heavy machine. Therefore, when the assembly component **13** is attached or detached, the surrounding monitoring can be performed using the detection unit **20** disposed at a predetermined position without changing the attachment position of the detection unit **20**. When the assembly component **13** is lifted using the mast **123** and the rotating platform **12** is turned to move the assembly component **13**, the surrounding monitoring can be performed on the basis of the detection of the detection unit **20** similar to that during normal work. Since the detection unit **20** is attached to the rotating platform **12**, it is easy to monitor whether or not there is an abnormality in the surroundings when the rotating platform **12** turns. Since the detection unit **20** is not connected via the assembly component **13**, a situation does not occur in which an error occurs in the position and angle of the detection unit **20** depending on the attachment state of the assembly component **13**, and uniform surrounding monitoring that does not depend on the attachment state can be realized.

Moreover, according to the work machine **1** of Embodiment 1, the detection unit **20** is disposed on a lower surface of the main frame **124** or below the lower surface. Therefore, blind spots are less likely to occur behind the work machine **1**, and it is possible to detect objects (workers, obstacles, or the like) that pass relatively below the counterweight **132** when the rotating platform **12** turns. In a case where the height of an object is low, when the rotating platform **12** turns, the object interferes with the rotating platform **12** or the rotating platform **12** (counterweight **132**) passes above the object. Accordingly, a determination as to whether or not the object interferes with the rotating platform **12** can be performed on the basis of the detection result of the detection unit **20**.

Moreover, according to the work machine **1** of Embodiment 1, the detection unit **20** is disposed so as to overlap the mast **123** when viewed vertically from above. Therefore, a single detection unit **20** can detect a region that extends to the left and right. Moreover, even when the detection unit **20** is at a position overlapping the mast **123**, since the detection unit **20** is attached to the lower portion of the main frame **124**, there is no risk of interference between the mast **123** and the detection unit **20**.

Moreover, according to the work machine **1** of Embodiment 1, the detection unit **20** is disposed inside the turning radius RT. Therefore, on the basis of the detection result of the detection unit **20**, whether or not there is an abnormality from the inside of the turning radius RT to the vicinity and outside of the turning radius RT can be monitored. Moreover, according to the work machine **1** of Embodiment 1, since the counterweight **132** overlaps the detection range W**1** of the detection unit **20**, the turning radius RT, that is, the rear end portion of the rotating platform **12** can be directly detected when the rotating platform **12** turns. Therefore, whether or not there is an abnormality in the vicinity of the turning radius RT can be monitored with higher accuracy.

Embodiment 2

FIG. **7** is a side view showing a work machine according to Embodiment 2 of the present invention when a first lifting capacity is set. FIG. **8** is a side view showing the work machine according to Embodiment 2 when a second lifting capacity lower than the first lifting capacity is set. A work machine **1**A according to Embodiment 2 is configured similar to the work machine **1** according to Embodiment 1 except that the attachment position of the detection unit **20** for the surrounding monitoring is different.

In Embodiment 2, the detection unit **20** is attached to a weight below the counterweight **132** in the uppermost stage mounted on the work machine body **10** when the second lifting capacity (for example, the lowest lifting capacity) lower than the first lifting capacity is set. In an example of FIGS. **7** and **8**, the counterweights **132** mounted at the second lifting capacity are the base weight **132**A and two-stage first weights **132**B, which are also attached to the work machine body **10** even when other lifting capacities are set. In this case, the detection unit **20** may be attached to a lower portion of the base weight **132**A, the side surface or rear part of the base weight **132**A or of the first weight **132**B, or the like. Additionally, in the case of a configuration in which the first weight **132**B is suspended below the base weight **132**A, the detection unit **20** may be attached to a lower portion of the first weight **132**B positioned in the lowermost stage.

According to such an attachment position of the detection unit **20**, even when the work machine **1**A is set to have any lifting capacity, the detection unit **20** is disposed at a constant position and angle, and thus constant monitoring processing can always be performed during work.

Moreover, as shown in FIGS. **7** and **8**, the detection unit **20** is attached to a lower portion of the counterweight **132**. Thus, a blind spot is less likely to occur behind the work machine **1**A, and objects (workers or obstacles) located below the counterweight **132** can also be detected when the rotating platform **12** turns. In a case where the height of an object is low, when the rotating platform **12** turns, the object interferes with the counterweight **132** or the counterweight **132** passes above the object. Accordingly, a determination as to whether or not the object interferes with the counterweight **132** can be performed on the basis of the detection result of the detection unit **20**.

Moreover, the detection unit **20** is disposed at a position overlapping the mast **123** when viewed from above. The position overlapping the mast **123** when viewed from above may be a position overlapping the strut **123***a* or the horizontal member **123***b* when the mast **123** is at any angle from a horizontal angle to a vertical angle. By virtue of such a disposition, a single detection unit **20** can detect a region that extends to the left and right. Moreover, even when the detection unit **20** is at a position overlapping the mast **123**, since the detection unit **20** is attached to the lower portion of the counterweight **132**, there is no risk of interference between the mast **123** and the detection unit **20**.

Moreover, the detection unit **20** is disposed inside the turning radius RT of the rotating platform **12**. According to such a disposition, the monitoring from the inside of the turning radius RT to the vicinity and outside of the turning radius RT is possible. Moreover, the detection range W**1** of the detection unit **20** overlaps the counterweight **132** (for example, a rear end of a lower surface of the base weight **132**A). According to such a disposition, the turning radius RT, that is, the rear end portion of the counterweight **132** can be directly detected when the rotating platform **12** turns, or the like, and whether or not there is an abnormality in the vicinity of the turning radius RT can be monitored with higher accuracy.

Moreover, a recessed portion U**1** (see FIGS. **5**A and **5**B) is provided at the lower portion of the base weight **132**A, and the detection unit **20** is attached such that at least a part thereof enters the recessed portion U**1**. In a case where the detection unit **20** is attached to the side surface or rear part of the base weight **132**A or of the first weight **132**B, recessed portions may be provided in these spots of the weights **132**A and **132**B, and part or all of the detection unit **20** may be accommodated in the recessed portions. According to such an attachment structure, interference of the detection unit **20** attached to the base weight **132**A with other configurations or surrounding objects, or an increase in the turning radius can be suppressed.

As described above, according to the work machine **1**A of Embodiment 2, even when the number of stages of the counterweight **132** is changed to switch the setting of the lifting capacity, the position of the detection unit **20** does not change. Thus, uniform monitoring processing can be realized.

The respective embodiments of the present invention have been described above. However, the present invention is not limited to the above embodiments. For example, in the above embodiments, a crawler crane has been shown as the mobile crane, but the movement method of the mobile crane according to the present invention is not limited, cranes adopting any movement methods, such as truck cranes or wheel cranes, may be used, and the type of cranes may be any cranes such as tower cranes. Additionally, in the above embodiments, a crawler crane has been shown as the work machine that is assembled or disassembled at a work site, but the work machine according to the present invention is not limited to a crane, and any work machines such as hydraulic excavators, may be used as long as the work machines are assembled and disassembled at the work site. Additionally, in the above embodiments, a scanning range finder and an imaging camera have been used as an example of the detection unit. However, the detection unit may have any configuration as long as the detection unit acquires data that enables the detection of surrounding objects. In addition, the details illustrated in the embodiments can be appropriately changed without departing from the scope of the invention.

The present invention is applicable to a work machine and a mobile crane.

It should be understood that the invention is not limited to the above-described embodiment, but may be modified into various forms on the basis of the spirit of the invention. Additionally, the modifications are included in the scope of the invention.

What is claimed is:

1. A mobile crane that is assembled or disassembled at a work site, the mobile crane comprising:

a crane body; and

a counterweight that includes weights in a plurality of stages and that is attached to the crane body,

wherein a setting is capable of being changed to any one of a plurality of lifting capacities by changing the number of stages of the weights attached to the crane body,

the plurality of lifting capacities includes a first lifting capacity and a second lifting capacity that is a lower lifting capacity than the first lifting capacity,

the mobile crane further comprises a detection unit that monitors surroundings attached to the counterweight, and

the detection unit is attached to any one of the weights other than an uppermost stage of the weights attached when the second lifting capacity is set.

2. The mobile crane according to claim **1**,

wherein the counterweight includes a base weight that supports another weight, and

the detection unit is attached to a lower portion of the base weight.

3. The mobile crane according to claim **1**,

wherein the detection unit is disposed below the counterweight.

4. The mobile crane according to claim **1**, further comprising:

a mast,

wherein the detection unit is disposed at a position being overlapped by the mast when viewed vertically from above.

5. The mobile crane according to claim **1**,

wherein the counterweight includes a recessed portion that accommodates the detection unit.

6. The mobile crane according to claim **1**,

wherein the detection unit is disposed inside a turning radius.

7. The mobile crane according to claim **1**,

wherein a part of the counterweight is included in a detection range of the detection unit.

8. The mobile crane according to claim **1**,

wherein the detection unit is disposed rearward of a frontmost portion of the counterweight.

9. The mobile crane according to claim **8**,

wherein the detection unit is disposed at a position overlapping the counterweight when viewed vertically from below.

10. The mobile crane according to claim **8**,

wherein the detection unit is disposed at a side part or a rear part of the counterweight.

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